MILITARY SPECIFICATION

CANISTERS, CHEMICAL PROJECTILE: 8 INCH, XM27 AND XM26, COMPONENTS FOR

This specification is approved for use by US Army Armament Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers components for two types of canisters.

1.2 Classification. The canister components shall be furnished in the following types as specified (see 6.2).

Type I - XM27 Canister Components.
Type II - XM28 Canister Components.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

BB-N-411 - Nitrogen, Technical.
BB-H-1168 - Helium, Technical.

FSC 1320

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, ATTN: DRDAR-TSC-S, Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter. :

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:01Z
Check the source to verify that this is the current version before use.

QQ-S-781 - Strapping, Steel, Flat and Seals.
PPP-F-320 - Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade) and Cut Shapes.

MILITARY

MIL-P-15011 - Pallets, Material Handling, Hardwood, Post Construction, 4-Way.
MIL-A-2550 - Ammunition, General Specification For.

STANDARDS

FEDERAL

Fed-Std-123 - Marking for Domestic Shipment (Civil Agencies).

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-1168 - Lot Numbering of Ammunition.

DRAWINGS AND TECHNICAL DATA PACKAGE LIST (TDPL)

US ARMY ARMAMENT RESEARCH AND DEVELOPMENT COMMAND

CHEMICAL SYSTEMS LABORATORY

TDPL15-12-163 - Canister, Chemical Projectile: 8 inch, XM27, Components For.
TDPL15-12-157 - Canister, Chemical Projectile: 8 inch, XM28, Components For.

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing D15-12-163 and Drawing D15-12-157 or on the subsidiary drawings shall conform to the specifications listed thereon or to the specific characteristics set forth on the drawings.

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:01Z
Check the source to verify that this is the current version before use.

3.1.2 Components. All components of the canisters shall conform to the specifications and drawings listed on Drawing D15-12-163 and Drawing D15-12-157 and subsidiary drawings.

3.2 Burst disk. Mechanical properties of the burst disk shall be in accordance with the requirements of Drawing B15-12-159 (see 4.1.2).

3.3 Assembly. The canisters shall be assembled in accordance with Drawing D15-12-163 or D15-12-157. The following special procedures and precautions shall be taken:

(a) Within 4 hours before welding, all component parts shall be cleaned of grease, oil, rust or other foreign material.

(b) The component parts of both ends of each canister shall be welded to the body within 4 hours of each other. If the assembly becomes contaminated it shall be recleaned before proceeding.

(c) Leakage testing of the assembled canister (4.4.4.2) shall be accomplished within 4 hours after final cleaning or welding. Immediately thereafter the interiors of the canisters shall be filled or purged for 15 seconds with dry nitrogen, type 1, class 2, grade B of BB-N-411, at a minimum flow rate of 4 cubic feet per minute (cfm) for type I canisters and 2 cfm for type II canisters, and the fill ports covered in accordance with Drawing D15-12-157 and D15-12-163.

(d) Within 4 hours after leakage testing and purging, surface treatment shall be applied to all exterior surfaces.

(e) During and after the above operations, special precautions shall be taken to avoid contamination and rusting, including the wearing of clean gloves during handling.

NOTE: If more than 4 hours elapses or is expected to elapse between cleaning and welding or welding and painting, the components or assemblies shall be examined, recleaned and protected.

3.4 Functioning. The canister, type I, burst disk shall withstand a minimum internal hydrostatic pressure of 250 pounds per square inch gage (psig) and shall shear against the inside edge of the ring in accordance with figure 1 with a maximum internal hydrostatic pressure of 450 psig, applied at a rate of 10 to 30 psig per second when tested as specified in 4.4.4.1. The canister, type II, burst disk shall withstand a minimum external hydrostatic pressure of 250 psig and shall shear against the inside edge of the canister in accordance with figure 1 with a maximum external hydrostatic pressure of 450 psig applied at a rate of 10 to 30 psig per second, when tested as specified in 4.4.4.1.

Source: https://assist.dla.mil -- Downloaded: 20260-22T21:01Z
Check the source to verify that this is the current version before use.

3.5 Leakage. The assembled canisters before painting, shall not leak helium at a rate greater than 1.0 X 10^{-6} atmospheric cubic centimeters per second, under a vacuum of 15 microns of mercury when tested as specified in 4.4.4.2.

3.6 Preproduction. Prior to the start of regular production, a preproduction lot shall be produced in accordance with this specification for examination and test (see 4.3).

3.7 Workmanship. The components shall be free from contamination (for eign matter, rust, oil or grease) and damage such as dents, cracks or deformation which could affect their intended use and shall conform to the standards of workmanship for parts and asemblies listed in MIL-A-2550. Special handling shall ensure canister interior remains clean and that burst disk exterior surface is clean prior to painting.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Contractor's responsibility. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.2 Objective evidence. The contractor shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspections.

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Lot. A preproduction lot of 100 canisters of the type specified (see 6.2) shall be manufactured using the same methods, materials, equip ment, processes and procedures as will be used during regular production.

Source: https://assist.dla.mil -- Downloaded: 20260-22T21:01Z
Check the source to verify that this is the current version before use.

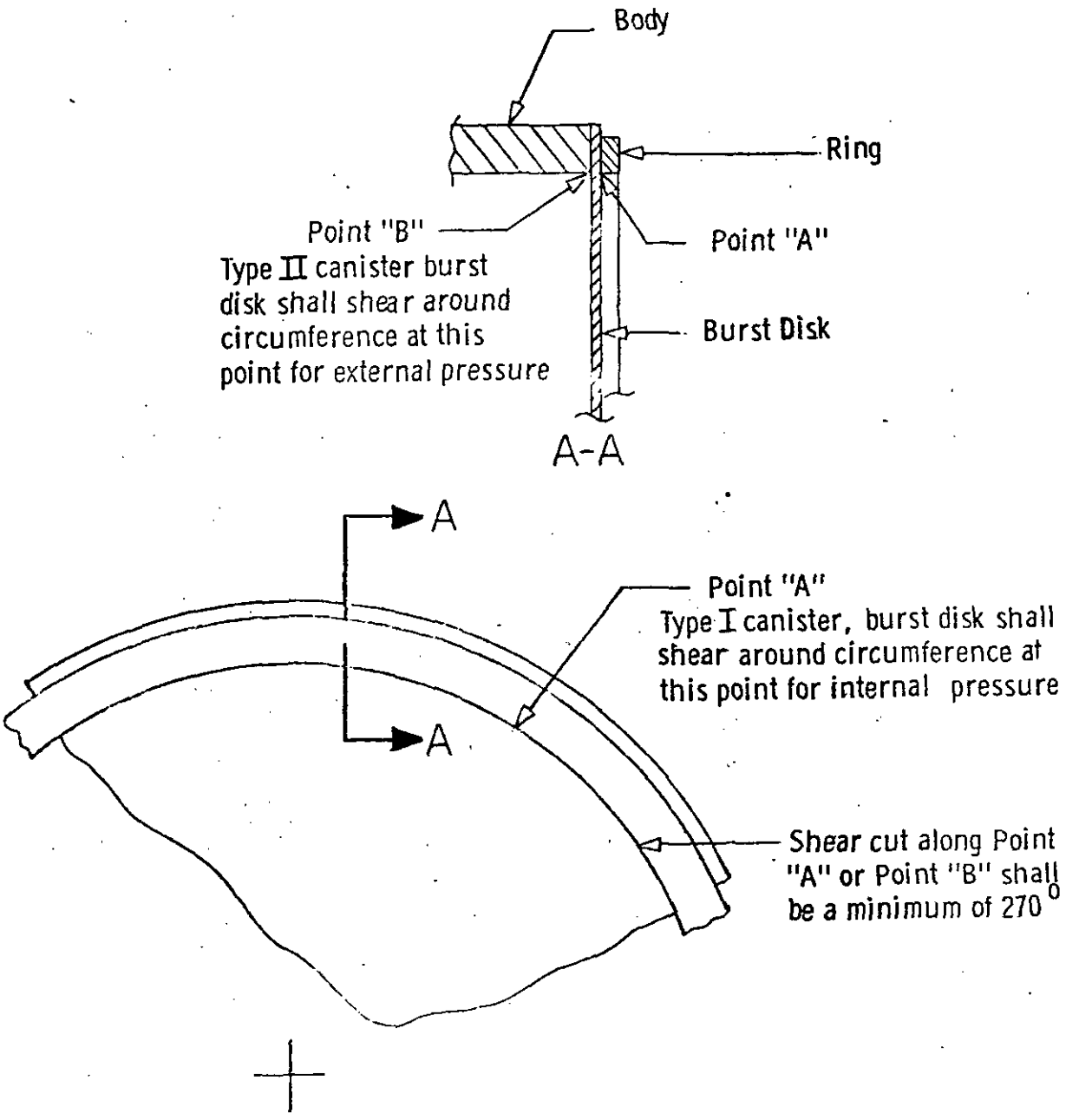


Figure 1
Burst disk and
ring functioning

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

4.3.2 Sampling.

4.3.2.1 For examination and nondestructive test. Sampling of each type specified shall be conducted in accordance with MIL-STD-105 (see 6.2).

4.3.2.2 For destructive test. Five canisters of each type (see 6.2) shall be taken at random from the preproduction lot.

4.3.3 Inspection procedure.

4.3.3.1 For examination and nondestructive test. The sample canisters shall be examined and tested in accordance with the classification of defects (4.4.3.4) and MIL-STD-105.

4.3.3.2 For destructive test. The sample canisters shall be tested in accordance with 4.4.4.1. The acceptance number is zero.

4.3.4 Acceptance/rejection criteria. The preproduction samples shall comply with 4.3.3 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

4.4 Quality conformance inspection.

4.4.1 Lotting.

(a) Canisters. Each lot shall be identified and controlled in accordance with MIL-STD-1168. A lot shall consist of canisters of one type, one interfix number, produced by one manufacturer, from the same materials and under essentially the same manufacturing conditions. No lot shall contain more than one burst disk lot.

(b) Burst disk. Each burst disk lot shall contain steel strip for burst disk fabrication from not more than one heat of steel.

4.4.2 Sampling.

4.4.2.1 For examination and nondestructive test. Sampling of each type specified shall be conducted in accordance with MIL-STD-105 (see 6.2).

4.4.2.2 For destructive test. Sampling of each type specified shall be conducted in accordance with MIL-STD-105, inspection level S-2 (see 6.2).

4.4.3 Inspection procedure.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:01Z
Check the source to verify that this is the currentversion before use.

4.4.3.1 For examination and nondestructive test. The sample canisters shall be examined in accordance with the classification of defects (4.4.3.4) and MIL-STD-105.

4.4.3.2 For destructive test. Sample canisters shall be tested in accordance with 4.4.4.1 using and AQL of 2.5 percent defective.

4.4.3.3 For critical defects. Each item in the lot shall be inspected for the critical characteristics listed in the classification of defects.

4.4.3.4 Classification of defects.

(a) Ring (Drawing C15-12-158).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
	101	Inside diameter edge
	102	Marking missing or incorrect
Minor:	AQL 2.5 percent defective	
	201	Outside diameter
	202	Inside diameter
	203	Position of diameters
	204	Thickness
	205	Flatness
	206	Deformation of flat surface

(b) Burst disk (Drawing B15-12-159).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
	101	Material thickness
	102	Surface imperfection
Minor:	AQL 2.5 percent defective	
	201	Diameter

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:01Z
Check the source to verify that this is the current version before use.

4.4.3.4 Classification of defects (continued).

(c) Weld plug (Drawing B15-12-180).

Categories		Defects
Critical:		None defined
Major:		None defined
Minor:	AQL 2.5 percent defective	
	201	Plug diameter
	202	Plug thickness
	203	Sharp edges unbroken

(d) Canister cup, long (Drawing D15-12-206).

Categories		Defects
Critical:		None defined
Major:	AQL 1.0 percent defective	
	101	Outside diameter
	102	Inside diameter
	103	Runout of diameters
	104	Perpendicularity of open end
	105	Length to basic diameter
	106	Inside length
	107	Inside diameter corner radius or chamfer of open end
	108	Slot position, depth and width
Minor:	AQL 2.5 percent defective	
	201	Maximum length
	202	Hole diameter
	203	Position of hole
	204	Diameter of undercut after welding
	205	Position of undercut
	206	Finish of outside diameter
	207	Perpendicularity of closed end
	208	Finish of open end
	209	Expulsion plate thickness
	210	Slot length
	211	Workmanship (3.7)

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

4.4.3.4 Classification of defects (continued).

(e) Canister cup, long, inertia welded (Drawing D15-12-232).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter
102	Inside diameter
103	Runout of diameters
104	Perpendicularity of open end
105	Length to basic diameter
106	Inside length
107	Inside diameter corner radius of open end
108	Slot position, depth and width
Minor:	AQL 2.5 percent defective
201	Maximum length
202	Hole diameter
203	Position of hole
204	Diameter of undercut
205	Position of undercut
206	Finish of outside diameter
207	Perpendicularity of closed end
208	Finish of open end
209	Expulsion plate thickness
210	Slot length
211	Workmanship (3.7)

(f) Canister cup, short (Drawing D15-12-207.

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective.
101	Outside diameter after welding
102	Inside diameter
103	Runout of diameter
104	Perpendicularity of open end
105	Perpendicularity of closed end

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:01Z
Check the source to verify that this is the current version before use.

4.4.3.4 Classification of defects (continued).

(f) Canister cup, chort (Drawing D15-12-207) (continued).

Categories	Defects
Major:	AQL 1.0 percent defective
106	Inside length
107	Inside diameter chamfer and counterbore of open end
108	Overall length
109	Base plate thickness
110	Position of blind hole
111	Blind hole diameter
112	Blind hole depth
Minor:	AQL 2.5 percent defective
201	Finish of open end
202	Finish of outside diameter
203	Hole diameter
204	Position of hole
205	Workmanship (3.7)

(g) Canister cup, short, inertia welded (Drawing D15-12-231).

Categories	Defects
Critical:	None defined
Major:	AQL 1.0 percent defective
101	Outside diameter after welding
102	Inside diameter
103	Runout of diameter
104	Perpendicularity of open end
105	Perpendicularity of closed end
106	Inside length
107	Inside diameter chamfer and counterbore of open end
108	Overall length
109	Baseplate thickness
110	Position of blind hole
111	Blind hole diameter
112	Blind hole depth

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:01Z
Check the source to verify that this is the current version before use.

4.4.3.4 Classification of defects (continued)

(g) Canister cup, short, inertia welded (Drawing D15-12-231) (Continued).

Categories	Defects
Minor:	AQL 2.5 percent defective
201	Finish of open end
202	Finish of outside diameter
203	Hole diameter
204	Position of hole
205	Workmanship (3.7)

(h) Canisters, chemical projectile: 8 inch, XM27 and XM28, components for (Drawing D15-12-163 and D15-12-157).

Categories	Defects	Acceptance standards
Critical:		
1	Helium leakage	4.4.4.2
2	burst disk damage	
3	Orientation of ring	
Major:	AQL 1.0 percent defective	
101	Overall length	
102	Perpendicularity of ring end	
103	Position of ring	
104	Maximum diameter	
105	Vinyl tape seal	
Minor:	AQL 2.5 percent defective	
201	Paint inadequate or damaged	
202	Workmanship (3.7)	

4.4.4 Tests. Tests shall be conducted as follows:

4.4.4.1 Functioning.

(a) Type I. The canister assembly shall be placed in a suitable holding fixture which will securely clamp the ring face in a manner restricting the ring from deforming outward without restricting rupturing of

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

burst disk. The required hydrostatic pressure shall be applied to the canister interior through the hole in the heavy plate end to determine the pressure of shear (see 6.3).

(b) Type II. The canister assembly shall be placed in a suitable holding fixture which will permit a hydrostatic pressure to be applied to the external face of the burst disk. The required hydrostatic pressure shall be applied to determine the pressure of shear (see 6.3).

4.4.4.2 Leakage test. The canister assembly shall be placed on a test head manifold connected to a vacuum system and a mass spectrometer type helium leak detector. The canister shall be enclosed in a suitable envelope and helium introduced into the envelope to approximately atmospheric pressure. The test head shall be evacuated to the specified level. The leak detector shall be calibrated to respond to a leak of 1×10^{-6} atmospheric cubic centimeters per second of helium when a standard leak is introduced at that point.

5. PACKAGING

5.1 Packaging, Type I Canisters, (Interplant shipment). The empty canisters shall be fitted with a protective closure conforming to Caplug model B526 or equivalent. The protective closure shall be placed securely over the burst disk end of the canister to provide protection. The canisters shall then be bound together in groups of ten in a manner to resemble a pyramid and banded with two strips of steel strapping conforming to QQ-S-781, flat, 3/4 x .023, type 1, class B, per bundle. Strips of fiberboard, conforming to grade W5c of PPP-F-320, shall be used as required to separate one canister from the other in the bundle and shall also provide a cushion on which the banding shall ride so as not to mar the painted surface of the canisters.

5.2 Packaging, type II canisters, (interplant shipment). The empty canisters shall be fitted with a protective closure conforming to Caplug, model B526 or equivalent. The protective closure shall be placed securely over the burst disk end of the canister to provide protection.

5.3 Palletization, type I canisters, (interplant shipment). Two bundles of canisters (20 canisters) shall be placed together, burst disk end to burst disk end, on a wooden pallet conforming to style 1, type 1, class 1 of MIL-P-15011, wood preservative treatment is not required. The bundles shall then be strapped to the pallet using flat steel strapping conforming to QQ-S-781, 1 1/4 x .035, type 1, class B in a manner to assure that the load will not shift during normal transportation.

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.

5.4 Palletization, type II canisters, (interplant shipment). Empty canisters shall be palletized vertically in one layer on a pallet conforming to style 1, type 1, class 1 of MIL-P-15011. Interlocking fiberboard separators shall be constructed and placed in a manner to separate the canisters one from the other. Side boards consisting of nominal 2 x 4 lumber shall be secured around the periphery of the pallet to secure the load and maintain its location on the pallet during shipment. In addition the load shall be secured with one horizontal steel band applied around the canisters, conforming to QQ-S-781, type 1, class B, 1-1/4 wide by .035 thick, inches.

5.5 Marking. In addition to any special marking required by the contract or order, marking of the palletized load shall be in accordance with FED-STD-123.

6. NOTES

6.1 Intended use. These metal parts assemblies are intended as components of the XM27 (type I) and XM28 (type II) canisters to be used in the XM736 projectile.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Type of canister components required (Type I or Type II).
(c) Preproduction:

(1) Time allowed for supplier submission of sample for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time required for the Government to notify the supplier whether or not to proceed with production.

6.3 Caution. Due to hazards attending the testing of items with high pressures, suppliers are hereby advised to take all necessary precautions to protect personnel while testing.

6.4 Supersession data. This specification supersedes Edgewood Arsenal Purchase Description EA-C-1160, 15 February 1977.

Custodian:

Army - EA

Preparing activity:

Army - EA

Project Number 1320-A474

Source: https://assist.dla.mil -- Downloaded: 202610-22T21:01Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS: This form is provided to solicit beneficial comments which may improve this document and enhance its use. DoD contractors, government activities, manufacturers, vendors, or other prospective users of the document are invited to submit comments to the government. Fold on lines on reverse side, staple in corner, and send to preparing activity. Attach any pertinent data which may be of use in improving this document. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity. A response will be provided to the submitter, when name and address is provided, within 30 days indicating that the 1426 was received and when any appropriate action on it will be completed.

NOTE: This form shall not be used to submit requests for waivers, deviations or clarification of specification requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

DOCUMENT IDENTIFIER *(Number)* AND TITLE

MIL-C-51460

NAME OF ORGANIZATION AND ADDRESS OF SUBMITTER

☐ VENDOR ☐ USER ☐ MANUFACTURER

1. ☐ HAS ANY PART OF THE DOCUMENT CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE? ☐ IS ANY PART OF IT TOO RIGID, RESTRICTIVE, LOOSE OR AMBIGUOUS? PLEASE EXPLAIN BELOW.

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDED WORDING CHANGE

C. REASON FOR RECOMMENDED CHANGE(S)

2. REMARKS

SUBMITTED BY *(Printed or typed name and address — Optional)*

TELEPHONE NO.

DATE

DD FORM 1426 1 OCT 76

EDITION OF 1 JAN 72 WILL BE USED UNTIL EXHAUSTED.

Source: https://assist.dla.mil -- Downloaded: 202640-22T21:01Z
Check the source to verify that this is the current version before use.

FOLD

POSTAGE AND FEES PAID
DOD-314

OFFICIAL BUSINESS
PENALTY FOR PRIVATE USE $300

Commander
US Army Armament Research
and Development Command
ATTN: DRDAR-TSC-S
Aberdeen Proving Ground, MD 21010

FOLD

Source: https://assist.dla.mil -- Downloaded: 2026-10-22T21:01Z
Check the source to verify that this is the current version before use.